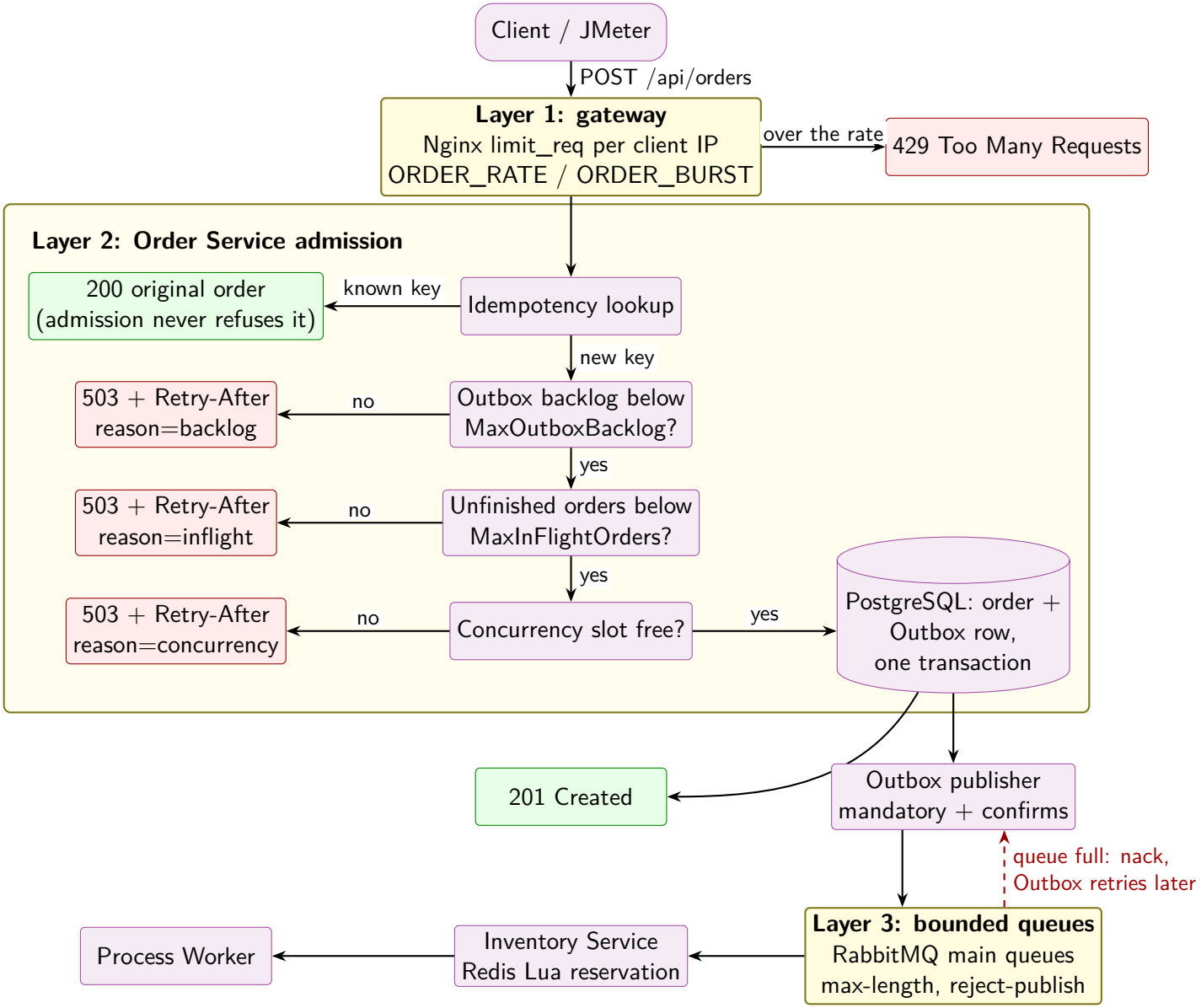

Client / JMeter
POST /api/orders
Layer 1: gateway
Nginx limit_req per client IP
ORDER_RATE / ORDER_BURST
over the rate
429 Too Many Requests
Layer 2: Order Service admission
200 original order
(admission never refuses it)
known key
Idempotency lookup
new key
503 + Retry-After
reason=backlog
no
Outbox backlog below
MaxOutboxBacklog?
yes
503 + Retry-After
reason=inflight
no
Unfinished orders below
MaxInFlightOrders?
yes
503 + Retry-After
reason=concurrency
no
Concurrency slot free?
yes
PostgreSQL: order +
Outbox row,
one transaction
201 Created
Outbox publisher
mandatory + confirms
queue full: nack,
Outbox retries later
Layer 3: bounded queues
RabbitMQ main queues
max-length, reject-publish
Inventory Service
Redis Lua reservation
Process Worker